

Human disturbance and the magnitude of phenotypic divergence

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1 Introduction

Human activities rapidly alter natural environments, and populations often respond with measurable phenotypic change. A central question in ecology and evolution is whether phenotypic divergence differs between human-disturbed and natural contexts, and whether these differences appear as larger average divergence, greater heterogeneity among systems, or both. Here, phenotypic divergence refers to the extent to which trait values differ between biologically relevant reference and comparison groups, such as earlier and later samples, or populations sampled under different environmental conditions. Such divergence may involve morphology, life-history traits, behaviour, phenology, or physiology, and may vary across disturbance categories such as climate change, harvesting, introductions, pollution, and landscape change.

Early comparative work asked whether human activities accelerate rates of phenotypic change relative to more natural contexts (Hendry et al., 2008). More recent work using an expanded comparative dataset found that phenotypic change was often greater in human-disturbed than in natural systems, and that some disturbance categories were associated with larger average responses whereas others showed a broader spread of outcomes across systems (Sanderson et al., 2022). It also found that some disturbance categories were associated with larger average responses, whereas others showed a broader spread of outcomes across systems (Sanderson et al., 2022). These findings suggest that human disturbance is an important correlate of phenotypic divergence, but they also raise two related questions that motivate the present study. First, how should divergence be measured when the interest is in the magnitude, rather than the direction, of phenotypic change? Second, do disturbance contexts differ only in average divergence, or also in the among-system variation of that divergence?

Previous studies have approached the first question using absolute Darwin or Haldane numerators (Sanderson et al., 2022). These numerators quantify the magnitude of phenotypic change before standardising by elapsed time, with Haldane numerators also scaling change by phenotypic variation (Gingerich, 1993; Haldane, 1949; Lynch, 1990). They are therefore closely related to the question addressed here: what is the magnitude of the difference between two groups relative to background variation? However, treating signed estimates as unsigned magnitudes also creates statistical challenges. Once the sign is removed, negative and positive values are folded onto the same positive scale. Similar issues arise whenever signed biological estimates are converted to unsigned magnitudes before analysis, including meta-analyses of assortative mating and phenotypic selection (Jiang et al., 2013; Morrissey, 2016). This folding can inflate small effects, impose a lower bound at zero, and generate non-Gaussian sampling distributions that are less compatible with standard

meta-analytic models (Nakagawa et al., 2026). These limitations matter when the goal is to compare the magnitude of phenotypic difference across disturbance contexts.

A second issue is whether phenotypic divergence differs not only in its average magnitude but also in its variability across systems. Two disturbance categories could show similar average divergence while differing markedly in how tightly estimates of divergence are distributed around that average. In such cases, focusing only on the mean would overlook biologically important differences in the consistency or predictability of phenotypic divergence. Modelling this heterogeneity is also statistically important because unmodelled differences in residual variation can affect inference about average patterns.

To address these issues, we will extend previous analyses using the Phenotypic Rates of Change Evolutionary and Ecological Database (PROCEED) that compiles studies of intraspecific quantitative phenotypic change across many taxa, traits, and disturbance contexts (Gotanda and Gorné, 2026). We will use $\ln M$ to quantify the magnitude of phenotypic divergence relative to within-group variation (Nakagawa et al., 2026), and location-scale meta-analysis to model both the mean and residual variation of that divergence across biological and methodological contexts (Nakagawa et al., 2025a).

2 Objectives and predictions

The objective of this project is to test whether phenotypic divergence differs between human-disturbed and natural contexts, and whether such differences are expressed in the average magnitude of divergence, the among-system variation in divergence, or both. We will estimate phenotypic divergence using $\ln M$, which measures the magnitude of the difference between two groups relative to within-group phenotypic variation. We will then use multilevel phylogenetic location-scale meta-regression to ask whether both mean divergence and residual heterogeneity in divergence vary across biological and methodological contexts.

We expect contrasts from human-disturbed contexts to show larger average $\ln M$ than contrasts from natural contexts. Biologically, this would mean that the groups defining each contrast differ more strongly from one another relative to the amount of phenotypic variation within those groups. We also expect some contexts to differ more clearly in the variation of divergence than in the average magnitude of divergence. For example, two disturbance categories could show similar average $\ln M$, but one could show tightly clustered estimates, whereas the other could show a broader range of divergence across systems. Such a pattern would indicate that contexts differ not only in the expected magnitude of phenotypic divergence, but also in the consistency

or predictability of that divergence.

Because $\ln M$ measures the amount of phenotypic divergence rather than the rate of change, elapsed time will be treated as a biological moderator rather than as part of the effect-size denominator. We will also account for the comparison structure of PROCEED, including the distinction between allochronic and syn-chronic contrasts, because these designs address related but non-identical biological questions. The candidate moderators and comparison structure are described below.

3 Methods

3.1 Data source and inclusion criteria

We will use the most recent version of the Phenotypic Rates of Change Evolutionary and Ecological Database (PROCEED; currently v6.2) together with the summary statistics needed for two-group comparisons and a broad set of biological and methodological descriptors. This project will focus on contrasts relevant to contemporary phenotypic change in human-disturbed and natural environmental contexts.

To be included in the main analysis, contrasts must contain the information required to estimate $\ln M$: `mean1`, `mean2`, `sd1`, `sd2`, `n1`, and `n2`. To remain comparable with earlier PROCEED analyses, we will restrict the primary dataset to contrasts spanning ≤ 300 generations when generation time is available. Contrasts with unknown generation time cannot be evaluated against this filter. We will therefore report how many contrasts are excluded for this reason.

We will retain the disturbance categories used in the recent reanalysis and current PROCEED structure, including climate change, hunting/harvesting, introduction, landscape change, response to introductions, other, and pollution. Candidate biological and methodological moderators are listed in Table ???. These include disturbance context, comparison design, elapsed time, trait type, taxonomic group, study type, environmental-change context, data type, and transformation status. The final model structure and sensitivity analyses will depend on data coverage across these variables.

3.2 Comparison structure

PROCEED contains two broad comparison designs. In allochronic studies, sample 1 is the earlier sample and sample 2 is the later sample. These contrasts follow a population or study system through time and therefore provide an estimate of phenotypic change along a temporal trajectory. In synchronic studies, the two samples represent different populations or conditions sampled after a known divergence interval. These contrasts estimate divergence between groups, rather than the trajectory taken by either group through time.

This distinction is important because rates or magnitudes of divergence are not necessarily equivalent to rates or magnitudes of change. Two populations can become different from each other even if both have changed, and the amount of divergence between them can differ from the amount of change that occurred within either population over the same period (Hendry and Kinnison, 1999). In this project, $\ln M$ will be interpreted as a measure of between-group divergence for both designs. For allochronic contrasts, that divergence corresponds to change through time within a population or system. For synchronic contrasts, it corresponds to divergence between populations or conditions.

In some synchronic contrasts, sample 1 corresponds to the ancestral, source, or original-condition population, and sample 2 corresponds to the derived or new-condition population. However, this ancestral-to-derived interpretation does not apply to all synchronic contrasts. We will therefore use the PROCEED variable `samp1_anc` to identify cases where sample 1 can be interpreted as ancestral, source, or reference condition.

We will account for comparison design by using `design` as a candidate moderator and, where data coverage permits, by testing whether elapsed time relates differently to $\ln M$ in allochronic and synchronic contrasts. We will also report design-specific summaries so that temporal change and condition-based divergence can be interpreted separately when needed.

3.3 Variables used in the analysis

The reanalysis will use the core identifiers, comparison-structure variables, moderators, and two-group summary statistics recorded in PROCEED. Table 1 lists the identifiers, comparison variables, and summary statistics used to define each contrast and estimate $\ln M$. Table 2 lists the biological and methodological moderators used in the main models and sensitivity analyses.

Table 1: Core PROCEED identifiers, comparison-structure variables, and summary statistics used in the reanalysis.

Variable	Role	Definition / use in this study
es_ID	Identifier	Unique identifier for each individual contrast or rate entry. Used to identify the unit from which each $\ln M$ estimate is calculated.
ref_ID	Identifier	Unique identifier for the primary source of information, such as a paper, thesis, or report. Used to model study-level dependence.
sys_ID	Identifier	Unique identifier for each study system. A system is a population or set of populations of a given species experiencing a specific driver or disturbance. Used to describe repeated contrasts within systems and, in sensitivity analyses, to evaluate non-phylogenetic dependence among contrasts from the same system.
sp_ncbi	Identifier / biology	Species or subspecific taxon name standardized to NCBI taxonomy. Used for species-level structure and phylogenetic modelling.
design	Comparison structure	Comparison design. Allochronic refers to the same population or system sampled at different time points; Synchronic refers to different populations with a known divergence time. Used as a candidate moderator and for design-specific summaries.
sample1	Descriptive comparison variable	Identification of sample 1. In allochronic contrasts, sample 1 is the earlier sample. In synchronic contrasts, sample 1 is intended to represent the source population, original condition, or ancestral environment, but this interpretation will be checked using <code>samp1_anc</code> where relevant.
sample2	Descriptive comparison variable	Identification of sample 2. In allochronic contrasts, sample 2 is the later sample. In synchronic contrasts, sample 2 is the comparison population or condition, often representing the derived, introduced, polluted, invaded, or otherwise changed condition.
samp1_anc	Comparison structure	Indicates whether sample 1 can be interpreted as the ancestral population, source population, ancestral place, or original condition. Used to avoid assuming an ancestral-to-derived relationship in all synchronic contrasts.
years	Time variable	Elapsed time in years. In allochronic contrasts, this is the number of years between the start and end time points. In synchronic contrasts, this is the number of years from the divergence point to sampling.
gtime	Species attribute	Generation time in years, defined in PROCEED as the time in which a population replaces itself. Used to derive elapsed generations where available.
generations	Time variable / filter	Number of generations elapsed, calculated as <code>years/gtime</code> . Used for the ≤ 300 generations filter where generation time is available.
n1, n2	Summary statistics	Sample sizes for sample 1 and sample 2. Required for effect-size estimation.
mean1, mean2	Summary statistics	Trait means for sample 1 and sample 2. Required for effect-size estimation.
sd1, sd2	Summary statistics	Standard deviations around the trait means for sample 1 and sample 2. Required for effect-size estimation.

Table 2: Main PROCEED moderators and sensitivity variables used in the reanalysis.

Variable	Role	Definition / use in this study
taxa	Moderator	Coarse taxonomic group. Current PROCEED categories include annelid, mollusc, arthropod, fish, amphibian, mammal, bird, reptile, and plant.
genphen	Moderator	Study type. Phenotypic refers to measurements of individuals developed in nature. Genetic refers to common-garden studies or other quantitative-genetic approaches.
disturbance	Main moderator	Classification of the environmental change or driver associated with the comparison. Main levels include climate change, hunting/harvesting, introduction, pollution, landscape change, response to introductions, and other. In PROCEED, hunting/harvesting is coded as Hunt_harv .
env_change	Moderator	Environmental-change context. ongoing indicates environmental change occurring through time, with phenotypic change measured within that process. novel indicates environmental change with a defined starting point. This variable applies mainly to allochronic studies; synchronic contrasts are coded as novel in PROCEED.
trait_description	Descriptive / verification variable	Trait recorded in the words of the original authors, often including units or clarifying details. Used to verify trait identity and coding where needed, but not treated as a statistical moderator.
trait_type	Moderator	Broad trait category. Current PROCEED categories include phenology, behaviour, physiology, growth, size, other morphology, other life history, and response traits.
data_type	Moderator / sensitivity variable	Type of trait data or measurement form. Current PROCEED categories include linear, area, cube, count, proportion, time, date, temperature, rate, adimensional ratio, index, and other. Used as a candidate moderator or in sensitivity analyses where data coverage permits.
transf_data	Moderator / sensitivity variable	Transformation status of the trait values. Current PROCEED categories include raw, ordination scores, arcsine transformations, residual-based values, natural-log-transformed values, and log10-transformed values. Used to define inclusion rules and sensitivity analyses.
data_scale	Sensitivity variable	Measurement scale of the trait, classified as ratio , interval , or undefined. Used to describe data properties and inform sensitivity analyses.

3.4 Effect size and analytical framework

Our response variable will be $\ln M$, a variance-scaled measure of the magnitude of phenotypic divergence between two groups. Conceptually, the two-group version of $\ln M$ is closely related to the numerator of a haldane: both describe the difference between two group means relative to phenotypic variation. Unlike haldanes, however, $\ln M$ is not divided by elapsed generations. We therefore interpret $\ln M$ as the magnitude of phenotypic divergence, not as a rate of change.

For two groups with sample means $\bar{x}_1$ and $\bar{x}_2$, standard deviations s_1 and s_2 , and sample sizes n_1 and n_2 , we first define the between- and within-group mean squares as

$$MS_B = \frac{n_1 n_2}{n_1 + n_2} (\bar{x}_1 - \bar{x}_2)^2 \quad (1)$$

and

$$MS_W = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}. \quad (2)$$

The within-group variance is then

$$s_W^2 = MS_W, \quad (3)$$

and the between-group variance component is

$$s_B^2 = \frac{MS_B - MS_W}{n_0}, \quad n_0 = \frac{2n_1 n_2}{n_1 + n_2}, \quad (4)$$

where n_0 is the harmonic-mean sample size. The effect size is defined as

$$\ln M = \ln \left(\frac{s_B}{s_W} \right). \quad (5)$$

In this project, $\ln M$ represents the magnitude of phenotypic divergence between two groups relative to the amount of variation within groups (Nakagawa et al., 2026). Positive values indicate that the difference between groups is large relative to typical within-group variation, whereas negative values indicate that the difference between groups is small relative to within-group variation. Values near zero indicate that between-group divergence is similar in magnitude to the typical within-group standard deviation. Thus, $\ln M$ reflects how large the difference between groups is relative to background phenotypic variation, rather than whether the trait increased or decreased.

Because PROCEED records contrasts as summarized sample pairs rather than paired repeated observations on the same individuals, the primary analyses will use the independent-groups formulation of $\ln M$. Estimation will rely on SAFE, a simulation-based approach that generates bias-corrected point estimates and sampling variances for $\ln M$. This is important because the standard plug-in estimator can become unstable or undefined when the estimated between-group variation is small relative to within-group variation, whereas SAFE remains applicable in these cases (Nakagawa et al., 2025b).

3.5 Meta-regression of divergence

Let y_i denote the estimated $\ln M$ for contrast i , with sampling variance v_i estimated using SAFE (Nakagawa et al., 2025b). We model the mean magnitude of phenotypic divergence for contrast i as

$$y_i = \mu_i + u_i^{(\text{study})} + a_i + \epsilon_i, \quad (6)$$

where

$$\mu_i = \beta_0^{(l)} + \sum_{k=1}^p \beta_k^{(l)} x_{ki}. \quad (7)$$

Here, $\beta_0^{(l)}$ is the intercept for the location part of the model, x_{ki} are moderator values for contrast i , and $\beta_k^{(l)}$ represent the effects of those moderators on mean $\ln M$. The term $u_i^{(\text{study})}$ is the random effect associated with the primary source of information, and a_i is the phylogenetic effect structured by the species correlation matrix.

The residual term from Eq. 6 is modelled as

$$\epsilon_i \sim N(0, v_i + \sigma_i^2), \quad (8)$$

where v_i is the known sampling variance of y_i , and σ_i^2 represents the remaining heterogeneity in $\ln M$ after accounting for sampling error, moderators, study-level dependence, and phylogenetic structure.

We considered including both **ref_ID** and **sys_ID**, as well as the phylogenetic species-level effect, because these terms represent different sources of non-independence. Contrasts from the same primary source may share study design, measurement methods, reporting decisions, and extraction context. Contrasts from the same study system may also be non-independent because they come from the same population, location, disturbance history, or research programme. Finally, contrasts from related species may be non-independent because of shared ancestry. However, these grouping factors are strongly associated in PROCEED. Most systems correspond to a single species, most species are represented by a single system, and many systems are represented by a single primary source (Table 3). Including **ref_ID**, **sys_ID**, and the phylogenetic species-level effect together in the primary model may therefore make their variance components poorly identifiable. For this reason, the primary model will include study-level dependence through **ref_ID** and the phylogenetic species-level effect. We will evaluate models including **sys_ID** as a sensitivity analysis where data coverage and model diagnostics permit. If models including multiple highly associated grouping factors are weakly identifiable, we will report this and avoid interpreting their variance components separately.

To examine whether residual heterogeneity in $\ln M$ varies across contexts, we will extend the model to a location-scale meta-analytic framework (?). The scale model is written as

$$\log \sigma_i = \beta_0^{(s)} + \sum_{m=1}^q \beta_m^{(s)} z_{mi}, \quad (9)$$

where $\beta_0^{(s)}$ is the intercept for the scale part of the model, z_{mi} are moderators for contrast i , and $\beta_m^{(s)}$ represent the effects of those moderators on the residual standard deviation of $\ln M$. The parameter σ_i is not estimated in a previous model; it is estimated jointly with the location model as part of the location-scale meta-regression. Positive values of $\beta_m^{(s)}$ indicate greater residual variation in divergence, whereas negative values indicate less residual variation and therefore more tightly clustered, more consistent divergence.

Inference will focus on model estimates and their uncertainty rather than on binary significance thresholds. For each moderator, we will report estimated differences in mean $\ln M$, or in residual heterogeneity where relevant, together with uncertainty intervals. We will interpret these estimates in terms of their magnitude, direction, and biological meaning, rather than treating threshold-based statistical significance as the main criterion for support. This emphasis is important because very small effects can become statistically detectable in large datasets, whereas heterogeneous but biologically important effects may be difficult to summarize using binary significance decisions.

Together, these models allow us to ask whether the average magnitude of phenotypic divergence differs across disturbance and other biological or methodological contexts, and whether the residual variation in that divergence also differs across those contexts.

3.6 Phylogeny

To account for shared ancestry among species, we will build a phylogenetic scaffold derived from the Open Tree of Life ([Hinchliff et al., 2015](#)) using the `rot1` package (?). Species names will first be standardized to the taxonomy used in PROCEED and then matched to Open Tree taxonomy. We will generate an induced subtree for the analysed species, resolve unresolved nodes algorithmically, and assign branch lengths using Grafen’s method ([Grafen, 1989](#)). The resulting tree will be converted into a phylogenetic correlation matrix for use in the multilevel models.

This approach provides a unified and reproducible phylogenetic framework across the broad taxonomic range represented in PROCEED. It also allows the phylogeny to be tailored directly to the species included in the analysis. However, this tree should be regarded as a pragmatic comparative scaffold rather than a fully resolved, time-calibrated phylogeny. The topology is assembled from multiple sources, some relationships may remain uncertain, unresolved nodes are resolved algorithmically, and Grafen branch lengths do not correspond to empirically estimated divergence times. Accordingly, the phylogenetic component will be used to capture broad patterns of shared ancestry while acknowledging that it represents an approximate evolutionary history.

3.7 Sensitivity analyses

We will include a limited set of sensitivity analyses to evaluate whether the main conclusions are robust to key data and modelling decisions. First, we will assess the influence of small sample sizes by repeating the main analyses after restricting the dataset to contrasts with total sample size $n_1 + n_2 \geq 40$, following guidance from the $\ln M$ framework (Nakagawa et al., 2026). This analysis will show whether the main inferences depend on contrasts for which estimation of $\ln M$ may be less stable.

Second, we will evaluate the influence of trait measurement properties. PROCEED includes traits measured as linear dimensions, areas, volumes or masses, counts, proportions, dates, times, rates, ratios, indices, and other data types. These measurement forms may differ in how within-group variation is expressed and in how comparable $\ln M$ estimates are across traits. We will therefore examine whether results are sensitive to the inclusion of different `data_type` categories and, where data coverage permits, evaluate `data_type` as a moderator.

Third, we will assess the influence of transformation status. The primary analysis will use transformation categories considered appropriate for estimating $\ln M$, with particular attention to raw-scale and natural-log-transformed data. Other transformation categories, such as residual-based values, ordination scores, arcsine transformations, or log10-transformed values, will be evaluated in sensitivity analyses where sample size and interpretation permit.

Finally, we will evaluate the robustness of the random-effects structure. The primary model will include study-level dependence and a phylogenetic species-level effect. Because `sys_ID` and species identity are strongly associated in PROCEED, models including both `sys_ID` and the phylogenetic species effect may be weakly identifiable. We will therefore fit models including `sys_ID` as a sensitivity analysis where model diagnostics permit, and we will avoid interpreting variance components separately if the data do not support their independent estimation.

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A Additional tables

Table 3: Overlap among candidate random-effects grouping factors in the full PROCEED dataset.

Grouping factor	Comparison	Groups with only one match	Total groups	Only one match
Species	Systems per species	362	428	
System	Species per system	1800	1801	
Reference	Systems per reference	264	326	
System	References per system	1737	1801	

Note. “Only one match” indicates the number or percentage of levels in the grouping factor that correspond to exactly one unit of the second term in the comparison column. For example, under “Systems per species”, it denotes species represented in only one system.